Project 8 ML4T- Strategy Evaluation

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Introduction:

This paper will analysis two trading strategies: rule based Manual Strategy and a machine learning-based Strategy Learner. Both strategies will utilize the same set of technical indicators: MACD, RSI, Bollinger Band Percentage(BBP) and Momentum. For Manual strategy, a scoring mechanism is applied based on the indicators to create BUY or SELL signals. For Strategy Learner, Random Forest model is applied to learn the optimal trading patterns from historical data. Our hypothesis is that Both Strategy Learner and Manual Strategy will outperform the benchmark and Strategy Learner would be the better model to achieve an optimal portfolio performance.

In this paper, we will take stock JPM as our analysis target and evaluate both strategies across in-sample and out same period. Where in-sample is 2008.1.1 to 2009.12.31 and out-sample is 2010.1.1 to 2011.12.31. For each strategy, we are able to maximum buy 1000 shares or sell 1000 shares. The benchmark is buying 1000 shares on the first day and holding it without any action until the end. For both Manual Strategy and Strategy Learner, we will set Commission: $9.95, Impact: 0.005 as their default values.

Indicator Overview:

4 indicators are used in this project: MACD, RSI, BBP, Momentum.

MACD :

MACD = EMA(Fast Period) - EMA(Slow Period)

Fast period=12, slow period=26, signal period=9

usage:

In Manual strategy: Values >0.01 contribute +2 points to buy signals, values <-0.01 contribute +2 points to sell signals, with smaller contributions for less extreme values.

In Strategy Learner: MACD histogram values is direct input features without manual thresholding.

RSI :

RSI = 100 - (100 / (1 + RS))

RS = Average Gain / Average Loss over 14 periods

Relative Strength Index (RSI) takes a window time of 14 days to calculate for the average gain / loss.
Usage:
In Manual strategy: RSI values <30 contribute +2 points to buy signals, values >70 contribute +2 points to sell signals. Moderate values (30-40 and 60-70) contribute +1 point. In Strategy Learner: RSI values are fed directly into the Strategy Learner.

BBP **:**

$$BBP = (\text{Upper Band} - \text{Lower}) / (\text{BandPrice} - \text{Lower Band})$$

In this experiment:

Middle Band = 20-day SMA

Upper Band = Middle Band + (2 × 20-day Standard Deviation)

Lower Band = Middle Band - (2 × 20-day Standard Deviation)

Usage:
In Manual strategy: BBP values <0.2 contribute +2 points to buy signals, >0.8 contribute +2 points to sell signals.
In Strategy Learner: Raw BBP are fed directly into the Strategy Learner

Momentum **:**

$$\text{Momentum} = (\text{Current Price} / \text{Price n days ago}) - 1$$

In this experiment: we take 10-day lookback period, n = 10
Usage:
In Manual strategy: Values >0.02 contribute +1 point to buy signals, values <-0.02 contribute +1 point to sell signals.
In Strategy Learner: Momentum values serve as direct input to the Strategy Learner.

Manual Strategy:

I implement a score-based trading strategy allocating all 4 indicators and generate the trading decisions based on the score. The strategy will calculate each trading day's buy_sore and sell_score by evaluating the previous day's indicators's values. The calculation of buy and sell score are following below strategy:

MACD Histogram: Values > 0.01 add +2 to buy_score; values < -0.01 add +2 to sell_score
RSI: Values < 30 add +2 to buy_score; values > 70 add +2 to sell_score
Bollinger Band Percentage (BBP): Values < 0.2 add +2 to buy_score; values > 0.8 add +2 to sell_score
Momentum: Values > 0.02 add +1 to buy_score; values < -0.02 add +1 to sell_score

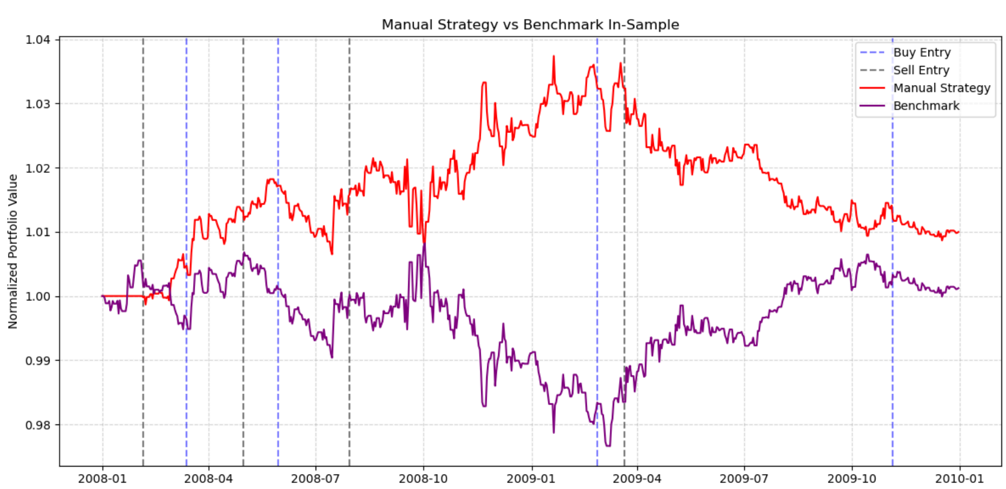
If either of the score reaches or exceeds 5 points, a buy/sell signal is generated. and the strategy maintains three possible positions:

Long: 1000 shares
Short: -1000 shares
Hold: 0 shares

Below is the action to take based on the score signal mechanism:

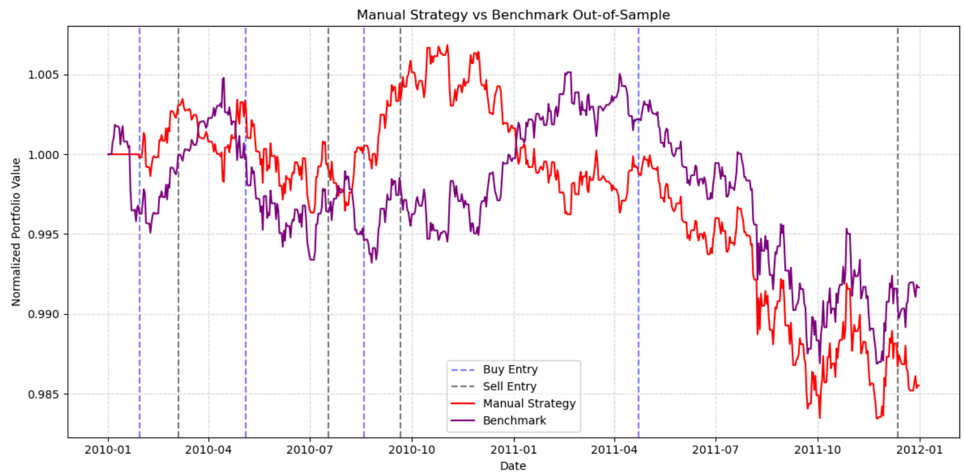
if position == -1000 and buy_signal: # Currently short
Buy 2000 shares → Long position (1000 shares)
elif position == 0: # No position
if buy_signal:
Buy 1000 shares → Long position (1000 shares)
elif sell_signal:
Sell 1000 shares → Short position (-1000 shares)
elif position == 1000 and sell_signal: # Currently long
Sell 2000 shares → Short position (-1000 shares)

Performance Analysis:



Manual Strategy In-Sample performance

In in-sample period, the above chart shows that Manual Strategy consistently outperforms the benchmark. The buy and sell entry point clearly indicates that the strategy can effectively identify the key entry and exit points. During the market downturns at 2008-2009, the strategy has successfully positioned itself to capture gains, and it reached the peak normalized value of around 1.04, while at the same time the benchmark is significant underperformance and it dropped to its lowest point, 0.98.



Manual Strategy Out-Sample performance

For the out-sample period, the same trading rules and parameters are used and here we got a different result. Manual Strategy has better performance in the first half, especially in Q3 and Q4 of 2010. Although the benchmark remains stable in this period, our strategy identified a few right movements and led the strategy performance into the peak at 1.008. However, in the downturn period after 2011.04, both benchmark and strategy keeps the same performance, and at the end manual strategy was slightly below the benchmark.

Performance Evaluation:

Metric	**In-Sample (2008-2009)**		**Out-of-Sample (2010-2011)**	
	Manual Strategy	Benchmark	Manual Strategy	Benchmark
Cumulative Return	0.00999	0.00123	-0.01449	-0.00834
STDEV of Daily Returns	0.00128	0.00134	0.00066	0.00067
Mean of Daily Returns	1.443e-05	2.580e-06	-1.980e-05	-1.126e-05

In-Sample and Out-Sample Performance Metrics

Above table is the metrics for manual strategy in both in-sample and out-sample. From the charts and matrics table, we could found that the manual strategy has outperfrmed the benchmark in in-sample period with a cumulative return of 0.999% compared to the benchmarks 0.123% and During the in-sample period, the strategy also have a better return with lower volatility: it has STDEV of 0.00128 vs benchmark's 0.00134, which also indicate that the strategy has a superior risk-adjusted performance.

In the out-sample period, both strategies gained a negative returns, and the Manual Strategy is slightly worse (-1.449% vs -0.834%). Since we are using the same manual strategy, this reveal that the market conditions has changed that the effectiveness of our scoring threshold established during in sample period is reduced.

The performance difference between in-sample and out-sample shows the challenges of keeping a consistent result between different market regimes. The 2008-2009 period has more pronounced trends with high volatility. While in 2010-2011, this periods shows a sideways movement wiht less persistent trends, the market movement makes it more difficult for the strategy to maintain its performance consistently.

Strategy Learner:

My strategy Learner implement the Random Forest model using Bag Learner containing multiple Random Tree(RTLearner) instances. Different from Manual strategy, Random Forest learners will optimal trading patterns directly from historical data with the same technical indicators.

Problem Framing:

This trading scenario is framed as a classification task by transforming historical price data into discrete trading signals:

Feature Engineering: The same four indicators as Manual Strategy will be used as input features. The combination of indicators (MACD, RSI, BBP, and Momentum) will be used to capturing different behaviors.

Target Variable Construction: We will set the target in three levels based on the future returns in 5 days:

- *Target = 1 (buy): If price increases by >(1.5% + impact) over next 5 days*
- *Target = -1 (sell): If price decreases by >(1.5%+ impact) over next 5 days*
- *Target = 0 (hold): Otherwise*

Training Methodology: The indicator features are shifted forward by one day to make sure the model is using the previous days' data to predict the next days action, in order to prevent look-ahead bias.

Hyperparameters and Implementation:

In this Random Forest model, a few key hyperparameters will be using. On the basic component of the model, Random Tree will take leaf_size = 10, to control when node becomes a leaf. On the top level, Bag Learner will take bags = 50 to create an ensemble of 50 Random Tree instances trained on bootstrapped samples.

The intention to use this RF model is to using its analysis power to discover complex relationships between indicators without relying on predefined thresholds. For example, when RSI values below 30, the Manual strategy might treat this as uniformly bullish, while the strategy Learner might identify this signal based on other indicators' combination and hold the action.

In the training process, the Baglearner will create 50 RTLearner instances and each trained on a different bootstrap sample of the training data. When making the prediction, each tree will "vote" in the action table and they will take the most common prediction (mode) as the final trading signal. The ensemble mechanism of many random bags will improve the robustness of the production and reduce the impact of any single tree's error.

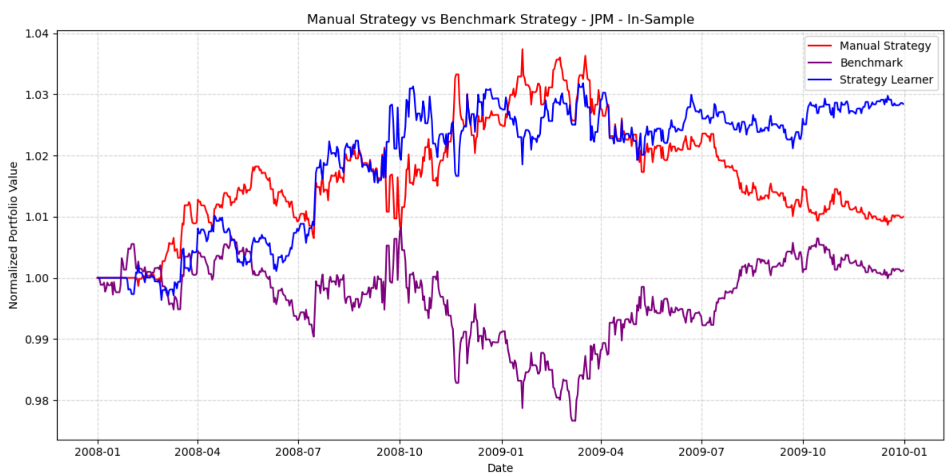
The RTLearner will recursively builds decision trees and randomly split the points at each node. A large set of trees will collectively capture the complex patterns in data. This random selection process combining with begging will help to prevent overfitting of the training data.

Experiment 1:

The Experiment 1 compares the performance of three trading approaches: Manual Strategy, Strategy Learner and Benchmark Strategy. Our hypothesis is that the Strategy Learner will outperform both Benchmark and Manual Strategy by identifying the complex patterns hiding in the indicator data matrix which predefined rules cannot identify well.

In this experiment we will keep the same in-sample and out-sample period and target stock JPM. Transaction costs are defined as: Commission = $9.95 per trade, Impact = 0.005. The position constraints is also the same as the previous setting: 1000 shares long, -1000 shares short, or 0 shares. The Strategy Learner will be trained in the in-sample period data and then be tested on both in-sample and out-sample period.

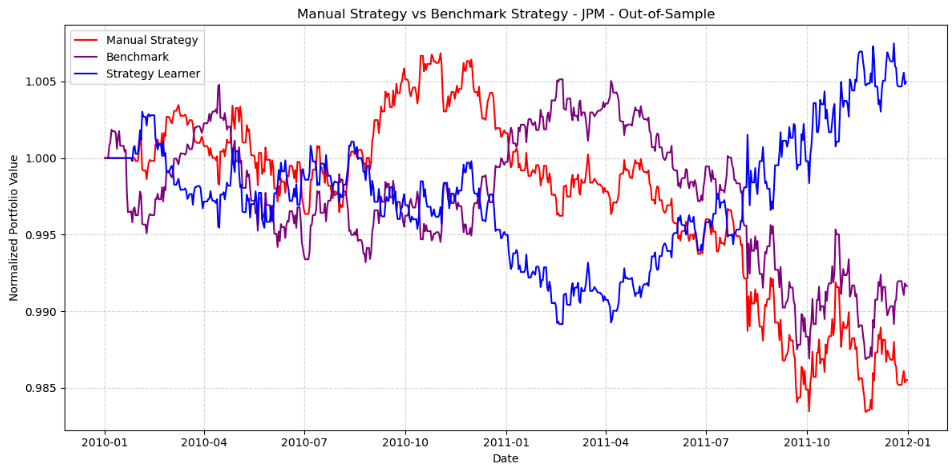
In-Sample Performance:



Experiment 1, In-Sample performance Result

The above in-sample result shows that both Manual and Strategy models has significantly outperform the benchmark. Strategy Learner achieved the highest normalized portfolio values of 1.029, while Manual strategy is 1.013 and benchmark is only 1. Through the whole training period, Strategy Learner maintained a more consistent growth, while the Manula strategy failed to identify the upturn period starting at Q2 2009 and started to decline.

Out-Sample Performance:



Experiment 1, Out-Sample performance Result

The above out-sample result shows a more dramatic divergence. The Strategy Learner has the best performance and provides a positive return despite the challenge market condition. Instead, the other strategy Manual Strategy suffered losses and the performance is very close to the Benchmark at the end of test period. Strategy achieved a 1.005 result while the other two line end with 0.985 and 0.991. From the chart we can tell that teh Strategy Learner's performance diverged significantly during the Q2 2011, where it reversed previous losses and the other strategies starting to decline.

Interpretation:

Strategy Learner outperform other strategies in both in-sample and out-sample tests. There might be a few reasons. First, Strategy Learner can discovers non-liner relationships between indicators and future return that human traders cannot recognize. Second, the Bag Learner approach can reduce overfitting by aggregating the predictions from multiple models, and making it more robust to market noise.

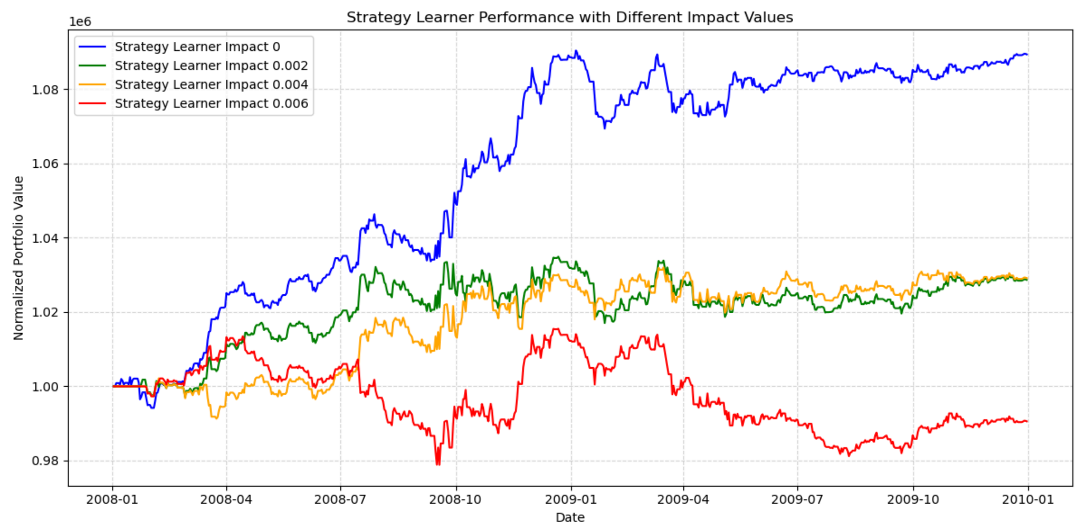
However, I don't expect that the Strategy learner will always beat benchmark everytime regardless test period and target stock. However, theoretically the general pattern of Strategy Learner could overperform Manual Strategy in most of time because of the fundamental advantage of machine learning approaches. The out-sample performance of Strategy Learner is inspiring beasuse it somehow reveal that the Strategy learner have learned the genuine patterns rather than simply overfitting into the training period.

Experiment 2:

In this experiment we will analysis the how the market impact cost will affect the Strategy Learner's performance. My hypothesis is that the market impact will have negative influence to the learner's performance. As the market impact cost increasing, the performance of Strategy Learner will getting worse.

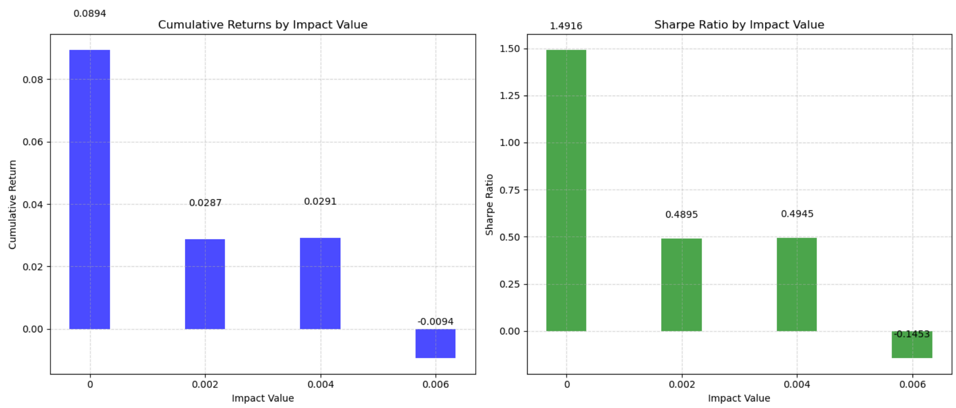
As systematic analysis is conducted to test this hypothesis. We will keep the same in-sample and out-sample period and target stock JPM. Transaction costs are defined as: Commission = $0 per trade. And the impact values tests will be: 0, 0.002, 0.004, 0.006. We will analysis two key metrics: Cumulative Return and Sharpe Ratio.

Results and Analysis



Experiment 2, Performance Result with different impact

Above charts reveals the performance of Strategy Learner under difference impact values. The 0 impact strategy dramatically outperform all others and achieving a nearly 9% cumulative return. However, the impact value with 0.002 and 0.004 has similar behavior and the 0.006 impact strategy actually loss money.



Experiment 2, Performance Metrics with different impact

The above bar charts reveal a clear quantitative perspective. For cumulative returns, we can see a striking drop between 0 and 0.002 impact. The returns remain similar at 0.004 impact (2.91%), but turn negative (-0.94%) at 0.006 impact. For Sharpe Ratio, it follows the similar change pattern as cumulative returns. The sharpe ratio drops 67% from 0 impact to 0.002 impact. and becomes negative at 0.006.

The above results confirm my hypothesis that a increasing impact cost will reduce the trading performance. One interesting finding is that the system is extremely sensitive to even small impact values (from 0 to 0.002). For future trading strategy development, these result emphasize the critical importance of adopting realistic transaction costs in the strategy design phase instead of adjusting them as an afterthought.